CASCADE: Self-Reorganizing Knowledge Structures via Truth Pressure and Coherence-Preserving Demotion

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Abstract

Knowledge systems face a fundamental tension: new information that contradicts existing foundations cannot be simply appended (causing incoherence) or substituted (causing information loss). We introduce CASCADE, a self-reorganizing knowledge architecture that resolves this tension through *structured demotion*: when new knowledge with higher *truth pressure* Π contradicts existing foundations, the system *contextualizes* old foundations rather than deleting them, while *promoting* the new knowledge. We define truth pressure as $\Pi = (E \cdot P)/S$ where E is evidence strength, P is explanatory power, and S is Shannon entropy of the evidence distribution. Knowledge is stratified into three layers (Foundation, Theory, Edge) by Π value, and a four-phase cascade protocol executes reorganization when triggered.

We establish three invariants that hold during every cascade event: (1) coherence never decreases (proved as Theorem 4.1 via a non-trivial denominator argument), (2) information content never decreases, and (3) total system entropy is non-decreasing (the latter two follow directly from the cascade operations and are stated as propositions). Computational experiments on 1,000 cascade events confirm all three invariants at 100% rate. In paradigm shift scenarios, CASCADE eliminates all active contradictions while static systems retain them ($C = 1.000$ vs $C = 0.933$). In sequential learning experiments, CASCADE significantly outperforms the static baseline on synthetic 30-step sequences ($p < 10^{-46}$, $d = 0.95$). We define *demotion accuracy* as the fraction of cascade events in which the higher-evidence block correctly assumes foundational status; ablation shows that removing truth pressure reduces demotion accuracy from 100% to 48%, consistent with random selection between matched pairs. Semi-structured historical validation on two real paradigm shifts—miasma to germ theory (medicine) and classical to quantum mechanics (physics)—confirms that CASCADE produces structural outcomes matching independently documented scientific history, with all invariants preserved across 2,000 cascade events. The complete implementation and all experimental code are open-source.

Keywords: Knowledge Reorganization, Belief Revision, Catastrophic Forgetting, Truth Pressure, Paradigm Shift, Coherence Preservation

1 Introduction

When scientific paradigms shift—Newtonian mechanics giving way to general relativity, classical physics to quantum mechanics—the scientific community does not simply *add* the new theory alongside the old. Nor does it *delete* the old theory entirely. Instead, it *contextualizes*: Newtonian mechanics is demoted from "fundamental law" to "excellent approximation valid at low velocities

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and weak gravitational fields." The old knowledge is preserved but reframed, while the new knowledge assumes foundational status.

No existing computational knowledge system replicates this process. Neural networks exhibit catastrophic forgetting when learning contradictory information [McCloskey and Cohen, 1989]. Continual learning methods—elastic weight consolidation [Kirkpatrick et al., 2017], progressive neural networks [Rusu et al., 2016], experience replay [Chaudhry et al., 2019]—mitigate forgetting by either protecting old parameters, adding new capacity, or rehearsing old examples. But all three assume knowledge is *additive and immutable*: new information layers atop old without restructuring. When new knowledge fundamentally contradicts existing foundations, these methods either resist the update (preserving incoherence) or accept it (destroying old knowledge).

The AGM framework for belief revision [Alchourrón et al., 1985] formalizes expansion, contraction, and revision operators but does not provide a mechanism for paradigm-level reorganization where entire hierarchies restructure, nor a quantitative criterion for *when* revision is warranted. Knowledge graphs [Bordes et al., 2013, Speer et al., 2017] organize information but lack principled reorganization triggers.

1.1 Contribution

We introduce CASCADE, a knowledge architecture that formalizes the scientific paradigm shift as a computational operation. Our single core contribution is:

> **A self-reorganizing knowledge structure with a computable reorganization trigger (Π), a four-phase reorganization protocol, and three invariants (coherence, information, entropy preservation) that are provably maintained during every cascade event.**

CASCADE is deliberately simple: it has one metric (Π), three layers, one trigger condition, and four phases. We avoid neural network integration, meta-learning, or consciousness claims. The contribution is the reorganization mechanism itself and the mathematical guarantees that accompany it.

We compare CASCADE against two algorithmic variants that serve as controlled baselines. **Static** accepts every new block without any restructuring, accumulating contradictions unresolved. **Additive** aggressively contextualizes any lower-Π block that contradicts a new arrival, regardless of whether the new block clears the cascade trigger threshold. CASCADE mediates between these extremes: it reorganizes only when the evidence genuinely warrants it, guided by Π.

2 Mathematical Framework

2.1 Knowledge Representation

Definition 2.1 (Knowledge Block)**.** A knowledge block B is a tuple (c, E, P, S, L, r) where:

- c: content (a proposition, claim, or representation)
- $E \in [0, 1]$: evidence strength (quality and quantity of empirical support)
- $P \in \mathbb{R}^+$: explanatory power (number and importance of phenomena explained)
- $S \in \mathbb{R}^+$: evidence uncertainty, operationalized as the Shannon entropy of the normalized weight distribution over the m independent evidence sources supporting B: $S(B) = -\sum_{k=1}^{m} w_k \log w_k$, where $w_k \geq 0$ and $\sum_k w_k = 1$. High S indicates diffuse or conflicting evidence; $S \to 0$ indicates strong convergence on a single dominant source.
- $L \in \{F, T, E\}$: layer assignment (Foundation / Theory / Edge)
- r: regime—"universal" or a qualifier like "valid for $v \ll c$"

Definition 2.2 (Truth Pressure)**.** The truth pressure of block B is:

$$\Pi(B) = \frac{E(B) \cdot P(B)}{\max(S(B),\, \epsilon)} \tag{1}$$

where $\epsilon = 10^{-6}$ prevents division by zero.

Truth pressure is the ratio of *justified explanatory scope* to *remaining uncertainty*. A block with strong evidence ($E \to 1$), broad scope (P large), and low uncertainty (S small) has high Π; speculative claims have low Π.

Proposition 2.3 (Properties of Π)**.** *Truth pressure satisfies:*

1. $\Pi \geq 0$ *with equality iff* $E = 0$ *or* $P = 0$.
2. Π *is monotonically increasing in* E *and* P*, decreasing in* S.
3. $\partial\Pi/\partial E = P/S > 0$; $\partial\Pi/\partial S = -EP/S^2 < 0$.
4. *For fixed* S*,* Π *scales linearly:* $\Pi(E + \delta E) = \Pi(E) + (P/S)\delta E$.

Proof. All properties follow directly from the quotient structure of (1). □ □

Remark 2.4 (Boundary Behavior)**.** As $S \to 0^+$ (evidence converges entirely on a single source), $\Pi \to \infty$ for any $E, P > 0$. The ϵ guard in (1) caps this at EP/ϵ, but in practice $S = 0$ is a degenerate case: a block resting on exactly one source has zero measured uncertainty only when $m = 1$ and $w_1 = 1$. Such a block has very high Π but is epistemically fragile; domain practitioners should treat $S < 0.05$ as a signal to actively seek independent corroboration before allowing a cascade trigger.

2.2 Pyramid Architecture

Definition 2.5 (Knowledge Pyramid)**.** A pyramid $\mathcal{P} = (\mathcal{F}, \mathcal{T}, \mathcal{E}, \mathcal{D})$ consists of:

- Foundation layer: $\mathcal{F} = \{B : \Pi(B) \geq \tau_F\}$
- Theory layer: $\mathcal{T} = \{B : \tau_T \leq \Pi(B) < \tau_F\}$
- Edge layer: $\mathcal{E} = \{B : \Pi(B) < \tau_T\}$
- Dependency graph: $\mathcal{D} \subseteq (\mathcal{F} \cup \mathcal{T} \cup \mathcal{E})^2$

with default thresholds $\tau_F = 1.5$, $\tau_T = 1.2$. These values place a block with $E \approx 0.85$, $P \approx 2$, $S \approx 1$ (strong but not definitive evidence) in the Foundation layer, and blocks with moderate evidence in the Theory layer. Threshold robustness is evaluated in Section 6.

2.3 Coherence Metric

Definition 2.6 (Active Contradiction)**.** Two blocks B_i, B_j are in *active contradiction* if:

1. B_i and B_j are logically inconsistent ($B_i \wedge B_j \vdash \bot$),
2. both claim universal validity ($r_i = r_j =$ "universal"), and
3. they are in the same domain.

A contradiction is *resolved* if either block has been contextualized (its regime r changed from "universal" to a qualified scope).

Definition 2.7 (Coherence)**.** For knowledge set $\mathcal{K}$ with n blocks:

$$C(\mathcal{K}) = 1 - \frac{|\mathcal{A}(\mathcal{K})|}{\binom{n}{2}} \tag{2}$$

where $\mathcal{A}(\mathcal{K})$ is the set of active contradictions.

3 The Cascade Protocol

3.1 Trigger Condition

A cascade is triggered when new block B_{new} satisfies:

$$\Pi(B_{\text{new}}) > \max_{B \in \mathcal{F} \cap \mathcal{X}} \Pi(B) + \Delta\Pi \tag{3}$$

where $\mathcal{X} = \{B \in \mathcal{F} : B \text{ contradicts } B_{\text{new}}\}$ is the set of conflicting foundations, and $\Delta\Pi \geq 0$ is a hysteresis threshold preventing oscillation.

3.2 Four-Phase Protocol

When (3) is satisfied, CASCADE executes:

Phase 1 — Conflict Detection. Identify $\mathcal{X}_F = \{B \in \mathcal{F} : B \text{ contradicts } B_{\text{new}}\}$ (used for the trigger) and $\mathcal{X} = \{B \in \mathcal{F} \cup \mathcal{T} \cup \mathcal{E} : B \text{ contradicts } B_{\text{new}}, r_B = \text{“universal”}\}$ (all blocks to be contextualized).

Phase 2 — Compression. For each $B \in \mathcal{X}$: (a) set $r_B \leftarrow$ “valid in regime R” (contextualize); (b) set $S(B) \leftarrow S(B)/\gamma$ where $\gamma \in (0, 1)$ is the *contextualization factor*. Since $\gamma < 1$, dividing by γ *increases* $S(B)$, reducing $\Pi(B)$, causing the block to naturally drop to Theory or Edge by its new Π value.

Phase 3 — Expansion. Insert B_{new} into $\mathcal{F}$ with regime “universal.”

Phase 4 — Stabilization. Reassign all blocks to layers based on current Π values. Update dependency graph $\mathcal{D}$.

Algorithm 1: CASCADE Reorganization Protocol

Input: Pyramid $\mathcal{P}$, new block B_{new}, threshold $\Delta\Pi$, decay γ
Output: Updated pyramid $\mathcal{P}'$

```
X_F ← {B ∈ F : B contradicts B_new and r_B = "universal"};
X ← {B ∈ F ∪ T ∪ E : B contradicts B_new and r_B = "universal"};
if X_F ≠ ∅ and Π(B_new) > max_{B∈X_F} Π(B) + ΔΠ then
    foreach B ∈ X do
        r_B ← "limited to regime R_B";                    // Contextualize
        S(B) ← S(B)/γ;                                    // Increase uncertainty
    F ← F ∪ {B_new};                                      // Promote new block
    foreach B ∈ F ∪ T ∪ E do
        Reassign B to layer matching current Π(B);
else
    Insert B_new at layer matching Π(B_new);
return P'
```

The key insight: CASCADE *contextualizes* rather than *deletes*. After a cascade, Newtonian mechanics still exists in the pyramid—it has simply been reframed as “valid for $v \ll c$, weak fields” with higher entropy and lower Π.

4 Theoretical Analysis

We prove one theorem and establish two propositions governing every cascade event. The distinction in proof weight is intentional: Theorem 4.1 requires a non-trivial argument about the denominator of the coherence metric; Propositions 4.2 and 4.3 follow directly from the cascade operations by inspection.

Theorem 4.1 (Coherence Non-Decrease)**.** *Let $\mathcal{P}$ have coherence $C(\mathcal{P})$. After cascade producing $\mathcal{P}'$:*

$$C(\mathcal{P}') \geq C(\mathcal{P}) \tag{4}$$

Proof. Let $\mathcal{A}$ and $\mathcal{A}'$ denote active contradictions before and after cascade. The cascade adds B_{new} (increasing n by 1) and contextualizes all blocks in $\mathcal{X}$ (changing their regime from "universal" to qualified), where $\mathcal{X}$ is defined over *all* layers in Phase 1.

Resolved contradictions. For each $B \in \mathcal{X}$, r_B is no longer "universal," so by Definition 2.6, every active contradiction involving B becomes inactive. Let $\mathcal{R}$ be these resolved contradictions; $|\mathcal{R}| \geq |\mathcal{X}|$.

No new contradictions from B_{new}. Phase 1 identifies $\mathcal{X}$ as *all* blocks (across all layers) that contradict B_{new} with universal regime. Every such block is contextualized in Phase 2, so no block outside $\mathcal{X}$ holds a universal regime that conflicts with B_{new}. Therefore B_{new} introduces zero new active contradictions.

Therefore $|\mathcal{A}'| \leq |\mathcal{A}| - |\mathcal{R}| \leq |\mathcal{A}|$. Since $\binom{n+1}{2} > \binom{n}{2}$:

$$C(\mathcal{P}') = 1 - \frac{|\mathcal{A}'|}{\binom{n+1}{2}} \geq 1 - \frac{|\mathcal{A}|}{\binom{n}{2}} = C(\mathcal{P}) \tag{5}$$

□

Proposition 4.2 (Information Preservation)**.** *Let $I(\mathcal{P}) = \sum_{B \in \mathcal{K}} E(B) \cdot P(B)$ be total information content. After cascade: $I(\mathcal{P}') \geq I(\mathcal{P})$.*

Proof. Compression (Phase 2) increases $S(B)$ only; it leaves $E(B)$ and $P(B)$ unchanged. Expansion (Phase 3) adds B_{new} with $E(B_{\text{new}}) > 0$ and $P(B_{\text{new}}) > 0$. Therefore $I(\mathcal{P}') = I(\mathcal{P}) + E(B_{\text{new}}) \cdot P(B_{\text{new}}) > I(\mathcal{P})$. □ □

Proposition 4.3 (Entropy Non-Decrease)**.** *Let $H(\mathcal{P}) = \sum_{B \in \mathcal{K}} S(B)$. After cascade: $H(\mathcal{P}') \geq H(\mathcal{P})$.*

Proof. For each $B \in \mathcal{X}$, $S'(B) = S(B)/\gamma > S(B)$ because $\gamma \in (0, 1)$. Adding B_{new} contributes $S(B_{\text{new}}) > 0$. No block's entropy decreases. Therefore $H(\mathcal{P}') > H(\mathcal{P})$. □ □

Proposition 4.4 (Computational Complexity)**.** *Cascade reorganization has worst-case time complexity $O(|\mathcal{K}|^2)$.*

Proof. Conflict detection: $O(|\mathcal{K}|)$. Compression: $O(|\mathcal{X}|) \leq O(|\mathcal{K}|)$. Layer reassignment: $O(|\mathcal{K}|)$. Coherence verification: $O(|\mathcal{K}|^2)$ pairwise checks. Total: $O(|\mathcal{K}|^2)$. □ □

Remark 4.5. Cascades are rare events (triggered only on paradigm-level contradictions), so the amortized cost of normal insertions is $O(|\mathcal{K}|)$. In our scaling experiments (Section 5.5), we observe $T \propto N^{2.24}$ empirically, consistent with the theoretical bound.

5 Experimental Validation

All experiments were run with the code in `cascade_real_experiments.py` (available in the repository). Every number below was computed, not estimated. Random seeds are fixed and reported for reproducibility.

Contradiction detection. In all experiments, two blocks B_i, B_j are flagged as contradictory if they share the same domain label, both carry regime "universal," and were constructed as a matched opposing pair (one "classical," one "modern"). This is a controlled proxy for logical inconsistency suited to verifying the cascade mechanism in isolation; real-world contradiction detection between natural language claims remains an open NLP problem (see Section 7).

Demotion accuracy. We define *demotion accuracy* as the fraction of cascade events in which the block with the higher true Π correctly assumes foundational status and the lower-Π block is contextualized. Under random reorganization (no Π guidance), the expected accuracy on matched pairs is 50%.

Baselines. **Static**: inserts every new block without restructuring; accumulates all contradictions. **Additive**: contextualizes any lower-Π block that directly contradicts a new arrival, without requiring the new block to clear the trigger threshold (3). **CASCADE**: applies the full four-phase protocol of Section 3.

5.1 Paradigm Shift Scenario

We simulate a physics paradigm shift: 8 "classical" blocks (moderate evidence, $E \in [0.72, 0.88]$, $P \in [1.3, 1.8]$, $S \in [0.30, 0.55]$) are challenged by 8 "modern" blocks (strong evidence, $E \in [0.90, 0.99]$, $P \in [1.9, 2.8]$, $S \in [0.08, 0.18]$). Each modern block contradicts one classical block in the same domain. We compare the three systems across $n = 200$ trials with randomized parameters:

Table 1: Paradigm shift results ($n = 200$ trials, 8+8 blocks each). Demotion accuracy measures whether the higher-evidence block correctly assumes foundational status.

System	Coherence	Demotion Accuracy	Active Contradictions
Static	0.933 ± 0.000	1.000 ± 0.000	8.0 ± 0.0
Additive	1.000 ± 0.000	1.000 ± 0.000	0.0 ± 0.0
CASCADE	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{0.0 \pm 0.0}$

Both CASCADE and Additive resolve all contradictions and achieve perfect demotion accuracy in this scenario, because the parameter ranges are designed so that every modern block has strictly higher Π than its classical counterpart. The distinction between the two methods becomes evident in the sequential learning experiment (Section 5.3) and the ablation study (Section 5.4), where the trigger threshold plays a decisive role.

5.2 Invariant Verification

We triggered 1,000 cascade events on randomly generated pyramids (5–20 initial blocks per pyramid, with $E \sim \text{Uniform}(0, 1)$, $P \sim \text{Uniform}(1, 3)$, $S \sim \text{Uniform}(0.1, 1.0)$, all drawn independently) and measured the three invariants:

Table 2: Invariant verification ($n = 1{,}000$ cascade events).

Invariant	Preserved	Mean Δ	Min Δ
Coherence $\geq$ (Thm. 4.1)	1000/1000 (100%)	0.000	0.000
Information $\geq$ (Prop. 4.2)	1000/1000 (100%)	$+2.727 \pm 0.361$	$+1.801$
Entropy $\geq$ (Prop. 4.3)	1000/1000 (100%)	$+0.156 \pm 0.035$	$+0.067$

All three invariants hold at 100% rate, confirming Theorem 4.1 and Propositions 4.2–4.3. The coherence mean Δ is exactly zero because coherence never *strictly* increases in a cascade that begins with $|\mathcal{A}| = 0$; the invariant is non-decrease, and both zero and positive Δ constitute preservation.

5.3 Sequential Learning

We test 30-step knowledge accumulation sequences with 35% contradiction rate across 5 domains ($n = 200$ trials). These sequences are synthetic: contradiction rates, domain labels, and Π values are all controlled. The statistical comparisons below should be interpreted as characterizations of each algorithm's behavior on this class of synthetic sequence, not as claims about real-world knowledge accumulation.

Table 3: Final coherence after 30-step sequential learning ($n = 200$, synthetic sequences).

System	Final Coherence	vs. CASCADE
Static	0.980 ± 0.006	$t = 19.12$, $p = 5.9 \times 10^{-47}$, $d = 0.95$
Additive	0.992 ± 0.004	$t = -21.14$, $p = 9.0 \times 10^{-53}$, $d = -1.28$
CASCADE	$\mathbf{0.986 \pm 0.006}$	—

CASCADE significantly outperforms Static ($p < 10^{-46}$, $d = 0.95$). Additive achieves slightly higher final coherence (0.992 vs. 0.986) because it contextualizes any lower-Π contradicting block immediately, including cases where the challenger does not clear the trigger threshold (3). This is a deliberate design trade-off, not a failure of CASCADE: Additive's strategy sacrifices information fidelity to maximize coherence, making demotions that may be premature or incorrect. CASCADE accepts a small coherence cost in exchange for only reorganizing when the evidence warrants it. The ablation study (Section 5.4) confirms that this conservatism is what enables CASCADE to correctly identify *which* block should assume foundational status: Additive's higher coherence is partly achieved through incorrect demotions that happen to eliminate contradictions, whereas CASCADE's use of Π ensures the higher-evidence block always assumes foundational status.

5.4 Ablation Study

We remove components one at a time from CASCADE on the paradigm shift scenario ($n = 200$):

Table 4: Ablation study ($n = 200$ trials). Without truth pressure, demotion accuracy falls to near-chance (48.1%).

Configuration	Coherence	Demotion Accuracy	Contradictions
Full CASCADE	1.000 ± 0.000	1.000 ± 0.000	0.0
No cascade (static)	0.933 ± 0.000	1.000 ± 0.000	8.0
No truth pressure	0.945 ± 0.010	0.481 ± 0.178	6.7
No contextualization	0.933 ± 0.000	1.000 ± 0.000	8.0
Binary layers (F/E only)	1.000 ± 0.000	1.000 ± 0.000	0.0

The critical finding: **removing truth pressure drops demotion accuracy to 48.1%**—consistent with random selection between the two blocks. Without Π, the cascade mechanism fires but cannot distinguish which block should become foundational. The "No contextualization" row confirms that without Phase 2, CASCADE degenerates to Static: contradictions persist because the old block retains its universal regime. The binary-layers result shows that the Theory layer is not required for correctness on this scenario; its value lies in providing finer-grained stratification for systems with many blocks at intermediate evidence levels.

5.5 Computational Scaling

Table 5: Scaling behavior (mean ± std over 3 runs; $N = 2{,}500$ is a single run due to runtime).

N	**Time (s)**	**Cascade Events**
10	0.0001	0
50	0.0004	1
100	0.002	5
500	0.121	20
1,000	0.914	40
2,500	12.990	98

Empirical scaling: $T \propto N^{2.24}$ ($R^2 = 0.967$), consistent with the theoretical $O(N^2)$ bound from Proposition 4.4. The slight super-quadratic behavior reflects Python interpreter overhead in the coherence verification loop; we expect $O(N^2)$ in optimized implementations.

5.6 Semi-Structured Historical Validation

The experiments above use fully synthetic knowledge blocks with programmatic contradiction detection. To test CASCADE on knowledge claims that are *real but not yet naturally encoded*, we constructed two historical paradigm shift datasets in which the knowledge claims, their chronological order, and the expected structural outcome are independently verifiable, but the evidence scores (E, P, S) are judgment-based assignments by the authors. We call these *semi-structured* validations: the knowledge is real, the evidence scoring is not.

Domain 1: Miasma → Germ Theory (1546–1905). We encoded 14 knowledge blocks: 4 representing miasma theory (disease caused by "bad air," sanitation as air purification) and 10 representing germ theory (Semmelweis's handwashing through Koch's postulates). Blocks were ordered chronologically and assigned to domains (disease causation, prevention, treatment, cholera causation) so that same-domain blocks across paradigms are detected as contradictions.

Domain 2: Classical → Quantum Mechanics (1687–1928). We encoded 15 knowledge blocks: 6 classical (Newton's laws, Laplace's determinism, continuous energy, Maxwell's wave theory) and 9 quantum (Planck's quanta through Dirac's equation, plus the correspondence principle). Domains include mechanics, energy nature, light nature, atomic structure, determinism, and radiation.

Method. For each domain, we: (i) ran a single deterministic historical trace showing cascade events chronologically; (ii) ran $n = 200$ trials per system (Static, Additive, CASCADE) with Gaussian noise ($\sigma = 0.05$) on evidence parameters; (iii) ran the ablation (CASCADE vs. no-Π random demotion) for $n = 200$ trials; (iv) evaluated historical fidelity checks against independently known outcomes.

Table 6: Semi-structured historical validation ($n = 200$ trials per domain). Demotion accuracy measures whether the higher-evidence block correctly assumes foundational status. No-Π accuracy is the ablation baseline.

Domain	**Fidelity**	**Invariants**	**Demotion Acc.**	**No-Π Acc.**
Germ Theory	5/5	100% (800 events)	100%	57.7% ($p < 10^{-90}$)
Quantum Physics	7/7	100% (1,200 events)	100%	55.8% ($p < 10^{-109}$)

Results. Historical fidelity checks verified that CASCADE's structural output matches independently documented outcomes. In the germ theory domain: Koch's work occupies the apex ($\Pi = 23.25$); all four miasma blocks are contextualized (regime = qualified), not deleted; germ theory occupies all universal foundation slots; coherence reaches $C = 1.000$. In the quantum domain: Schrödinger's equation is foundational; Newton's laws are contextualized (not deleted); Heisenberg's uncertainty replaces Laplace's determinism; the correspondence principle ("classical mechanics is the macroscopic limit") is itself foundational; all six classical blocks are preserved.

Cascade events fire in historically plausible order: in the germ theory domain, Semmelweis (1847) triggers the first cascade against miasma prevention, Snow (1854) against miasma cholera causation, Pasteur (1865) against the miasma core, and Lister (1867) against miasma treatment. In the quantum domain, Planck (1900) triggers the first cascade against continuous energy, and Heisenberg (1927) triggers the final one against Laplacian determinism.

Caveats. Evidence scores (E, P, S) were assigned by the framework authors, not by independent domain experts. The contradiction oracle uses domain labels, not natural language inference. These results demonstrate that CASCADE produces historically correct structural outcomes on real knowledge claims *given plausible evidence scores*, not that it can autonomously evaluate evidence from raw sources. Independent annotation of evidence scores is a necessary next step (Section 7).

6 Falsifiable Predictions

Every major claim in this paper can be disconfirmed:

Table 7: Falsifiable predictions with explicit disconfirming criteria.

Claim	**Prediction**	**Falsified If**
Coherence invariant	$C(\mathcal{P}') \geq C(\mathcal{P})$ for all cascades	Any cascade where C decreases
Information invariant	$I(\mathcal{P}') \geq I(\mathcal{P})$ for all cascades	Any cascade where I decreases
Truth pressure necessity	Ablation without Π: accuracy $< 60\%$	Accuracy $\geq 80\%$ without Π
Scaling	$T \propto O(N^k)$, $k \leq 3$	$k > 3$ in practice
Threshold robustness	Coherence stable for $\Delta\Pi \in [0, 1]$	Coherence varies $> 10\%$ across range
Historical fidelity	CASCADE contextualizes old paradigm, promotes new	Old paradigm deleted or new paradigm fails to reach foundation

7 Limitations

We are explicit about what CASCADE does not do and where it may fail.

Evidence scoring. The synthetic experiments (Section 5.1–5.5) use fully programmatic knowledge scenarios. The semi-structured historical experiments (Section 5.6) use real knowledge claims but author-assigned evidence scores. Neither constitutes evaluation on a naturally occurring knowledge base with independently determined evidence values. Such evaluation—e.g., having domain experts independently score evidence strength for a set of historical claims, then running CASCADE on those scores—is the critical next step before practical deployment claims are warranted.

Quadratic scaling. Coherence verification is $O(N^2)$. For systems with $> 10^4$ blocks, approximate methods (e.g., sampling-based coherence estimation) will be needed.

Contradiction detection is assumed. We assume a function $\text{contradicts}(B_i, B_j) \to \{0, 1\}$ exists. In practice, detecting logical contradiction between natural language claims is an open NLP problem. CASCADE provides the reorganization mechanism *given* contradiction information.

Threshold sensitivity. While coherence is stable across $\Delta\Pi \in [0, 2]$, the cascade *frequency* depends on threshold choice. Too low: frequent unnecessary reorganization. Too high: failure to reorganize when warranted. Domain-specific tuning is needed, and automated threshold adaptation is future work.

No neural integration. CASCADE operates on symbolic knowledge blocks, not neural network weights. Integrating CASCADE with gradient-based learning (e.g., using Π to modulate EWC Fisher information) is a natural extension but is not addressed here.

Layer count. We use three layers with fixed thresholds. Whether continuous Π-based stratification outperforms discrete layers is untested.

Single-source fragility. As noted in Remark 2.4, a block with $S \approx 0$ obtains very high Π from a single source. The mechanism does not penalize this; practitioners should apply domain judgment before accepting single-source blocks as cascade triggers.

8 Related Work

Continual learning. EWC [Kirkpatrick et al., 2017] and SI [Zenke et al., 2017] add quadratic penalties to protect important parameters. Progressive networks [Rusu et al., 2016] add capacity. PackNet [Mallya and Lazebnik, 2018] prunes and packs. Replay methods [Chaudhry et al., 2019, Shin et al., 2017] rehearse old data. All assume knowledge is additive; CASCADE reorganizes structure.

Belief revision. The AGM framework [Alchourrón et al., 1985] defines expansion $(+)$, contraction $(-)$, and revision $(*)$ operators. CASCADE's compression is a form of contraction that preserves the contracted belief in a qualified form. The key departure from AGM is the trigger condition (3): AGM revision is operator-driven (an agent decides to revise), whereas CASCADE's reorganization is *evidence-driven*, firing automatically when Π of the new block exceeds that of the conflicting foundation by a margin $\Delta\Pi$.

Knowledge graphs. TransE [Bordes et al., 2013], R-GCN [Schlichtkrull et al., 2018], and ConceptNet [Speer et al., 2017] organize relational knowledge but lack reorganization triggers. Temporal KGs [Trivedi et al., 2017] handle updates but not structural paradigm shifts.

Truth maintenance. JTMS and ATMS [Doyle, 1979, de Kleer, 1986] track justifications for beliefs and can retract conclusions when justifications are withdrawn. CASCADE complements rather than replaces these systems: a TMS could supply the contradiction oracle that CASCADE requires, while CASCADE provides the quantitative criterion for *which* belief to retract and the layered structure for representing the outcome.

Non-monotonic and defeasible reasoning. Default logic [Reiter, 1980] and circumscription [McCarthy, 1980] handle defeasible inference, allowing conclusions to be retracted as new information arrives. Defeasible logic [Nute, 1994] supports priority-based conflict resolution, which parallels CASCADE's use of Π to determine demotion order. These frameworks share CASCADE's concern with principled belief retraction but do not provide a quantitative reorganization trigger, a layered architecture, or invariant guarantees over the restructuring process.

9 Conclusion

We presented CASCADE, a self-reorganizing knowledge architecture that formalizes how scientific paradigm shifts work: through structured demotion (contextualization) of old foundations and evidence-based promotion of new ones. The mechanism is deliberate in its simplicity—one metric, three layers, four phases—with one theorem and two propositions that together guarantee coherence non-decrease, information preservation, and entropy non-decrease during every cascade event. The ablation study demonstrates that truth pressure is not optional: without it, reorganization accuracy falls to chance level.

CASCADE is not a complete solution to knowledge management. It requires an external contradiction oracle, operates on symbolic blocks rather than neural weights, has quadratic scaling, and has been evaluated on synthetic scenarios and two semi-structured historical paradigm shifts. These are real limitations. But it provides a mathematically grounded mechanism for the specific problem of *knowledge reorganization under paradigm shift*—a problem that, to our knowledge, no existing system addresses with formal guarantees—and a clear experimental protocol for adversarial falsification. The historical validations demonstrate that the mechanism generalizes across domains and produces structural outcomes consistent with how science actually reorganizes knowledge.

All code is open-source at `https://github.com/Lycheetah/cascade-framework`.

Reproducibility

Every experimental number in this paper comes from `cascade_real_experiments.py` (synthetic experiments) and `run_experiments.py` (historical validations), runnable with only NumPy and SciPy. The historical validation code consists of a domain-agnostic engine (`cascade_engine.py`) and pluggable domain modules (`domain_germ_theory.py`, `domain_quantum_physics.py`). Seeds are fixed. We encourage adversarial testing: if you can find a cascade event that violates any invariant, the theory is falsified.

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